

22-52-Q2

$$Q: f(x) = 0.5 + \frac{\sin^2 \left(\sum_{i=1}^3 \sqrt{(x_i - \delta)^2} \right) - 0.5}{1 + 0.001 \left(\sum_{i=1}^3 \sqrt{(x_i - \delta)^2} \right)^2}$$

$$\delta = 0$$

$$-10 \leq x_i \leq 10 \text{ for } i = 1, 2, 3$$

$$f_{\min}(x^*) = 0$$

$$x_i^* = 0$$

(a)

Chromosome ID	$x = (x_1, x_2, x_3)$	$f(x)$	Crossover Mate	Child 1	Child 2
1	$(-2.3, 1.5, 0.5)$		3		
2	$(1.1, -0.3, -5)$		4		
3	$(1.2, 0.8, -1.8)$		5		
4	$(7.8, 9.2, 3.3)$		1		
5	$(-0.5, 0.5, 0.8)$		2		

$$(b) \hat{z}_i = \alpha X_i + (1 - \alpha) y_i$$

$$\alpha = 0.4$$

(c) 18 bits

Solution (a) ① Table - 2

Chromosome ZD	$X=(X_1, X_2, X_3)$	$f(x)$	Crossover mate	Child 1	Child 2
1	$(-2.3, 1.5, 0.5)$	0.833	3	$(-0.2, 1.08, -0.88)$	$(-0.9, 1.22, -0.42)$
2	$(1.1, -0.3, -5)$	0.0327	4	$(5.12, 3.4, -0.02)$	$(3.78, 3.5, -1.68)$
3	$(1.2, 0.8, -1.8)$	0.376	5	$(0.18, 0.62, -0.24)$	$(0.52, 0.68, -0.76)$
4	$(7.8, 9.2, 3.3)$	0.844	1	$(1.74, 4.58, 1.62)$	$(3.76, 6.12, 2.18)$
5	$(-0.5, 0.5, 0.8)$	0.947	2	$(0.46, 0.02, -2.68)$	$(0.14, 0.18, -7.52)$

②

$$f(x) = 0.5 + \frac{\sin^2 \left(\sum_{i=1}^3 \sqrt{(X_i - \delta)^2} \right) - 0.5}{1 + 0.001 \left(\sum_{i=1}^3 \sqrt{(X_i - \delta)^2} \right)^2}$$

$$\delta = 0$$

$$-10 \leq X_i \leq 10 \text{ for } i = 1, 2, 3$$

$$f_{\min}(X^*) = 0$$

$$X_i^* = 0$$

$$f(x) = 0.5 + \frac{\sin^2 (|x_1| + |x_2| + |x_3|) - 0.5}{1 + 0.001 \times (|x_1| + |x_2| + |x_3|)^2}$$

ZD	$ x_1 + x_2 + x_3 $	$f(x)$
1	4.300	0.833
2	6.400	0.0327
3	3.800	0.376
4	20.300	0.844
5	1.800	0.947

-2.3, 1.5, 0.5

1.1, -0.3, -5

1.2, 0.8, -1.8

7.8, 9.2, 3.3

-0.5, 0.5, 0.8

(b) Arithmetic recombination ($\alpha = 0.4$)

$$\text{child 1} = 0.4x + 0.6y$$

$$\text{child 2} = 0.4y + 0.6x$$

(c) ①

$f_{\min}(X^*) = 0$, $X_i^* = 0$, so, we want to min $f(x)$
two fittest chromosomes is

Chr-ZD	(x_1, x_2, x_3)	18 bits
2	(1.1, -0.3, -5.0)	100011011111010000
3	(1.2, 0.8, -1.8)	100011100010011010

② Encoding : each variable is mapped from the range $[-10, 10]$ to a 6 bit integer

$$b = \text{round} \left(\frac{z-x}{y-x} (2^L - 1) \right)$$

$$= \text{round} \left(\frac{z+10}{20} \times 63 \right)$$

z	b	b bits
1.1	35	100011
-0.3	31	011111
-5	16	010000
1.2	35	100011
0.8	34	100010
-1.8	26	011010

(d) For $\delta = 0.32$

The optimum occurs at $x_i^* = 0.32$ ($i=1,2,3$)

$$b = \text{round} \left(\frac{0.32+10}{20} \times 63 \right) = 33 \rightarrow 100001$$

$$x^* = 10000/10000/10000/$$